

MAISHA RAHMAN CHOWDHURY

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IOWA STATE UNIVERSITY

Bachelor of Science, Computer Science, Minor in Applied AI
Magna Cum Laude, Dean's List

Aug 2022 – May 2026

GPA: 3.72

UNDERGRADUATE RESEARCH ASSISTANTSHIP

Adaptive Agentic Scheduling for Multi-Model LLM Inference | Dr. Asad W. Malik May 2026–Present

- Extending iGenOrch toward adaptive, learning-based orchestration for concurrent LLM inference under resource constraints.
- Developing a closed-loop LangGraph scheduling architecture that collects CPU, memory, and inference-latency telemetry and represents runtime conditions as system state.
- Investigating LinUCB contextual bandit scheduling to learn which OS-level intervention is most effective for different workload conditions.
- Comparing LinUCB, ϵ -greedy, and LLM-agent policies under average-latency and P90-latency objectives to evaluate adaptive and tail-latency-aware scheduling.
- Exploring dynamic latency-spike correction and token-budget adaptation to respond to changing inference performance.
- Current direction: extending workload-aware scheduling toward predictive and agentic orchestration that can anticipate resource contention and proactively adapt resource allocation.

Undergraduate Research Fellow, Summer iEDGE REU | Mississippi State University May–Aug 2025

- Developed a Linux kernel-aware scheduler on Raspberry Pi 5 to prioritize LLM tasks.
- Deployed TinyLLaMA and DeepSeek (1B) with automated startup and monitoring tools.
- Collected and analyzed latency, CPU, and memory metrics using Python; improved inference efficiency through dynamic scheduling using C.

Undergraduate Research Assistant | Iowa State University Jun–Aug 2024

- Researched the interpretability of code-trained LLMs.
- Built a Tkinter annotation tool to compare GPT-generated vs. human labels using Python.
- Identified error patterns (e.g., insufficient context, imprecision) in LLM labeling.

PUBLICATIONS

Authors: Faiza Akram, **Maisha Rahman Chowdhury**, Gabriella Dunnne, Katherine Williams, Kaleb Love, Asad Waqar Malik, Samee U. Khan, **Title:** iGenOrch: Intelligent Orchestration Framework for Multi-Model LLM Inference on Edge Platforms, ACMSE 2026

PRESENTATIONS

Undergraduate Research Showcase, Mississippi State University, Colvard Student Union

July 31, 2025

TECHNICAL SKILLS

- **Languages:** Python, Java, C/C++, JavaScript, TypeScript, SQL, Bash
- **Backend:** Node.js, Spring Boot, REST APIs, WebSocket
- **Frontend:** React, HTML, CSS, Tkinter
- **Databases:** MySQL, MariaDB, MongoDB, Neo4j, Firebase
- **Tools:** Git, GitLab CI/CD, Postman, Jupyter Notebook, Conda, LM Studio, LangGraph
- **Systems:** Linux, Bash/Unix, Raspberry Pi, Remote Server Development
- **AI:** LM Studio, LangGraph, Groq AI, Google API Key, n8n, Docker, LLM Evaluation, Machine Learning Fundamentals

PROJECTS

Throwly – Full-Stack Startup Application

Client: Elvis Kimara | Senior Design (COM S 4020)

Technologies: JavaScript, TypeScript, React, Node.js, Firebase, Git

- Implemented third-party identity verification integration to enhance platform trust and user authenticity.
- Developed and optimized item ranking and sorting algorithms based on user impressions and engagement metrics (most-clicked / popularity-based ordering).
- Designed and refined RESTful APIs in Node.js for authentication, validation, and feature expansion.
- Built modular, type-safe frontend components using React with TypeScript for scalable UI development.
- Conducted systematic testing and debugging to improve application reliability and performance.
- Managed version control using Git with feature branching, pull requests, and collaborative code reviews in an agile team environment.

Dungeon Crawler Game Engine

COM S 3270

Technologies: C, C++

- Implemented a terminal-based dungeon crawler game emphasizing low-level memory management and systems design.
- Designed procedural dungeon generation and grid-based navigation mechanics.
- Implemented four distinct monster AI behaviors.
- Applied data structures such as priority queues and grids for entity management.
- Engineered turn-based combat and state transition logic.

Clever Quest – Real-Time Multiplayer Game Platform

COM S 3090

Technologies: Java, Spring Boot, MySQL, MariaDB, WebSocket, Git

- Developed backend services for a real-time multiplayer gaming platform using Spring Boot.
- Implemented WebSocket-based bidirectional communication to support interactive games (Trivia, Guess the Number).
- Designed REST APIs for account creation, authentication, score storage, and profile updates.
- Designed relational database schemas and optimized SQL queries using MySQL and MariaDB.
- Implemented CI/CD workflows and version control using Git.
- Focused on concurrent client handling and real-time state synchronization.

Anime Merchandise E-Commerce Prototype

COM S 3190

Technologies: React, Node.js, HTML, CSS, JavaScript

- Developed a prototype full-stack e-commerce web application for anime merchandise sales.
- Built dynamic frontend interfaces in React, including cart management and checkout workflows.
- Implemented backend API endpoints using Node.js for inventory and order handling.
- Designed user authentication and session management features.
- Structured modular UI components with reusable state logic.
- Implemented full client-server architecture (prototype not deployed)

WORK EXPERIENCE

Teaching Assistant (TA) | Department of Computer Science, Iowa State University | Ames, IA August 26, 2024 – December 20, 2024

- Taught and supervised lab sessions to reinforce course material for ComS 127 taught in Python.
- Assisted students during office hours by clarifying concepts and reviewing assignments.
- Graded student work and provided constructive feedback.
- Collaborated with professors and other TAs to develop grading rubrics for exams.
- Answered student questions, ensuring a supportive and inclusive learning environment.

Math Grader | Department of Mathematics, Iowa State University | Ames, IA August 14, 2023 – May 11, 2024

- Graded assignments and exams with accuracy and consistency.
- Held office hours to help students understand mathematical concepts.
- Provided feedback to improve students' problem-solving and analytical skills.

Tutor | Academic Success Centre, Iowa State University | Ames, IA January 20, 2023 – April 01, 2023

- Tutored students individually in mathematics and computer science.
- Adapted teaching methods to different learning styles for improved comprehension.
- Supported students in exam preparation and guided them through problem-solving strategies.